

Name: _____

Date: _____

Class: _____

MyDesign Portfolio

Element K Initial Template

Element K: Reflection on the Design Project

Engineering Design Process Summary (K1, K2)

Summarize each step of the design process including:

- defining the problem (elements A-C)
- ideation (element D)
- selection & prototype design (elements D-F)
- prototype building (element G)
- testing (elements H-I)
- evaluation (element J)

For each project step, explain: What went well? What didn't go well? Why?

Lessons Learned About the Engineering Design Process (K3)

What do you think designers need to consider or do based on your experience? A level 5 reflection illustrates thorough and specific details on improving the design process in the future. Make sure to touch on lessons from across the whole engineering design process and experience that can apply to any future engineer. The lessons learned in Element K should focus on the engineering design process and how you implemented it. Later on (in L1) you will provide recommendations for continuing work on this project in particular.

Defining the Problem

Defining the problem went very well, we quickly figured out what we were trying to solve, including who the stakeholders were, and why it actually mattered.

Ideation/selection & prototype design

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The ideation process went very well, we came up with a lot of solid ideas pretty quickly. Once we came up with the ideas we moved to picking the best options, which we agreed upon as a group. From there we started to draw out a basic idea, adding in all our thoughts until we had a full plan to start prototyping.

Prototype Building

The prototype frame got built pretty quickly, with no issues. Once we started on the circuits it got a little harder. The actual circuit board went pretty seamlessly, but once we started trying to tie it into the motor it got a little worse. When tethering out wire we had to re-do it a couple times to get power, and a few other problems. After that we had to work on our actual collection system. Referring back to our previous plan we quickly constructed a basic shape for our net. After that we played around with a couple of collection devices eventually ending on an empty soda can. From there our collection system and proto-type as a whole was done.

Testing

The first test we did was in a small box to test buoyancy and it went pretty well, though we eventually discarded it because we drilled holes into our prototype to let water in. The second test was at the JCC where we tested buoyancy and maneuverability and it went pretty well. The last test was at Fort Mason, we played around with different configurations of pool floaties, and once we found an almost neutrally buoyant set up we tested its ability to dive and collect plastics. That went very well, we could go down and up easily, and collected some microplastics.